

GRAND MARAIS HARBOR, MN, USA

WMO: 997259

Lat: 47.7483N

Lon: 90.3415W

Elev: 611

StdP: 14.37

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -12.0	(c) -7.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 13.8	(c) 76.5	(d) N/A	(e) 73.1	(f) N/A	(g) 70.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(a) N/A	(b) N/A	(c) N/A	(d) WB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.8	14.8	15.8	27.7	37.8	46.9	53.9	61.9	62.3	56.3	45.4	32.3	20.2
	DBStd	18.55	12.74	10.39	10.11	6.87	5.07	5.08	5.52	4.39	5.94	6.59	9.54	10.53
	HDD50	4893	1091	958	692	368	123	17	0	0	14	175	532	923
	HDD65	9280	1556	1378	1157	815	562	335	128	108	265	607	981	1388
	CDD50	1152	0	0	1	2	26	135	370	381	204	33	0	0
	CDD65	64	0	0	0	0	0	3	33	23	5	0	0	0
	CDH74	219	0	0	0	0	1	25	113	71	9	0	0	0
	CDH80	27	0	0	0	0	0	5	10	11	1	0	0	0
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(15) Precipitation	PrecAvg	27.0	1.2	0.8	1.2	2.1	2.7	3.6	3.2	2.8	3.0	3.3	2.1	1.4
	PrecMax	35.9	2.5	1.9	3.3	7.6	4.8	8.4	8.0	5.6	6.4	9.0	3.9	3.0
	PrecMin	20.8	0.3	0.2	0.4	0.4	0.9	0.4	1.3	0.9	0.7	1.3	0.6	0.4
	PrecStd	4.3	0.6	0.5	0.7	1.5	1.0	1.7	1.7	1.1	1.4	1.9	1.0	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.6	38.5	53.0	60.5	67.2	77.9	81.5	80.6	74.4	65.7	53.7	41.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	37.1	35.3	47.2	54.5	61.9	69.6	77.5	75.4	70.4	60.3	50.1	38.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	34.9	32.9	43.4	50.1	58.1	65.8	74.0	72.0	67.5	57.6	47.5	37.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	32.6	30.6	40.0	46.8	54.9	62.4	71.1	69.4	65.1	55.1	45.3	35.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	14.5	14.4	13.0	12.4	13.8	14.9	15.0	13.8	12.1	10.7	10.8	12.4
		MCDBR	12.1	12.2	15.5	17.2	19.0	20.3	19.0	17.4	15.2	14.0	11.3	9.3
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.260	0.269	0.301	0.345	0.364	0.386	0.396	0.381	0.365	0.318	0.292	0.266
	taud		2.431	2.398	2.399	2.381	2.415	2.378	2.398	2.453	2.468	2.562	2.534	2.459
	Ebn at Noon		266	288	293	288	285	278	273	273	268	267	252	249
	Edh at Noon		20	26	31	35	36	37	36	33	29	23	19	18
(44) All-Sky Solar Radiation	RadAvg		380	696	1106	1489	1714	1842	1918	1629	1192	712	384	290
	RadStd		47	77	102	155	116	132	86	113	77	83	48	29

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page